

ON ORTHOGONAL BASES FOR RATIONAL LATTICES^{†)}

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We recall (see, for instance, [1, p. 19]) that the *lattice generated by vectors* $a_1, a_2, \dots, a_m$ in a Euclidean space V is the collection L of all vectors of V of the shape

$$\alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_m a_m, \quad \alpha_i \in \mathbb{Z}.$$

Henceforth we use function notations and denote this lattice L by $\mathcal{L}(a_1, \dots, a_m)$. Clearly, the members of a lattice form an additive subgroup of the group of all vectors of V .

Lattices appear in various fields of mathematics such as number theory, group theory (classification of simple finite groups and study of crystallographic groups), coding theory, the classical problem of sphere packing, etc. (see [2]). Lattices seem to appear for the first time in number theory, namely, in the section of it which is called "geometry of numbers" and whose foundations were laid in the works by Hermit, Minkowski, Voronoï, et al. (see [1; 3, Chapter 2]).

It is well known [4, p. 223] that every subspace U of a finite-dimensional Euclidean space V possesses an orthonormal basis. The natural question arises: given a finite-dimensional Euclidean space V and a lattice L in V , under what conditions does L possess an orthogonal basis in the sense of the standard inner product in V ? The present article addresses this question.

Fix some orthonormal basis for V and consider the coordinates of all vectors with respect to this basis. Call a lattice L in V *rational (integer)* if all vectors in L have rational (integer) coordinates. In this article we prove that construction of an orthogonal basis for a rational lattice reduces to construction of an orthogonal basis for some integer lattice. Afterwards we give a sufficient condition for existence of an orthogonal basis for an integer lattice. Next in §2 we examine two-dimensional integer lattices in more detail and give a criterion for existence of an orthogonal basis. In §3 we study complete two-dimensional lattices and give a simpler criterion for existence of an orthogonal basis which is based on construction of some canonical basis for a two-dimensional integer lattice. In §4 we use the results obtained to give a partial answer to Problem 9.45 in [5]: "Let a be a vector in $\mathbb{R}^n$ with rational components and let $S(a) = \{ka + b \mid k \in \mathbb{Z}, \text{ and } b \in \mathbb{Z}^n\}$. Find necessary and sufficient conditions in terms of the components of a for $S(a)$ to have an orthogonal basis in the sense of the standard inner product of $\mathbb{R}^n$." We give a sufficient condition for existence of an orthogonal basis for $S(a)$ in the case of $n > 1$, and obtain a sought criterion for $n = 2$.

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§ 1. A Sufficient Condition for Existence of an Orthogonal Basis

To begin, we introduce needed definitions and recall some known facts of use in the sequel. All definitions and notations are routine and can be found in [6, 7].

Fix a basis $e_1, \dots, e_n$, $n \geq 2$, for V . Then V is isomorphic to the direct sum of the subspaces $\mathbb{R}e_1 \oplus \mathbb{R}e_2 \oplus \dots \oplus \mathbb{R}e_n$. The space V contains the integer lattice $\mathcal{L}(e_1, \dots, e_n) \cong \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \dots \oplus \mathbb{Z}e_n$, and every other lattice is the image of this lattice under a suitable linear transformation of V . Each vector x in V can be written as a linear combination $x = x^1 e_1 + \dots + x^n e_n$, $x^i \in \mathbb{R}$. We write down the coordinates of x as a row $x = (x^1, \dots, x^n)$ and call x a *rational (integer)* vector if all coordinates

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$x^1, \dots, x^n$ are rational numbers (integers). If $y = (y^1, \dots, y^n)$ is another vector in V then the *inner product* of x and y is the number

$$x \cdot y = \sum_{i=1}^n x^i y^i.$$

By an *elementary transformation* of a matrix we always mean an integer transformation of the rows of the matrix, i.e., multiplication of some row by -1 , as well as multiplication of some row by an integer and addition to another row. If integer vectors $a_1, \dots, a_m$ generate a lattice L then we can find a basis for this lattice (there exists a basis since L is a torsion-free finitely-generated abelian group). Furthermore, with each system of generators $a_1, \dots, a_m$ of L we can associate the matrix $A = (a_i^j)$, $i = 1, \dots, m$, $j = 1, \dots, n$, whose rows are the coordinates of the vectors $a_1, \dots, a_m$. The converse is true as well: each integer matrix in $M_{m,n}(\mathbf{Z})$ corresponds to the lattice generated by the rows of the matrix. Using Euclid's algorithm and applying elementary transformations to the matrix A , we can reduce it to triangular form [8, p. 29]. Then the nonzero rows of the resultant matrix constitute a *ladder basis* for the lattice L :

$$\begin{aligned} c_1 &= c_1^k e_k + c_1^{k+1} e_{k+1} + \dots + c_1^n e_n, & c_1^k > 0, \\ c_2 &= c_2^l e_l + \dots + c_2^n e_n, & c_2^l > 0, \\ \dots & \dots \dots \dots \dots \dots \dots \dots \\ c_r &= c_r^s e_s + \dots + c_r^n e_n, & c_r^s > 0, \end{aligned} \quad (1)$$

where $c_i^j \in \mathbf{Z}$, $1 \leq k < l < \dots < s \leq n$. Hence, we can choose a ladder basis for each integer lattice.

It is well known [3, p. 119] that two linearly independent system of vectors $f_1, \dots, f_m$ and $g_1, \dots, g_m$ determine the same lattice if and only if they are connected by a unimodular transformation, i.e., when

$$g_i = \sum_{j=1}^m \alpha_{ij} f_j, \quad i = 1, \dots, m, \quad (2)$$

where $T = (\alpha_{ij})$ is an integer matrix with determinant ± 1 . The matrix $\Gamma = (f_i \cdot f_j)$ is called the *Gram matrix* of the basis $f_1, \dots, f_m$. Upon passage to a new basis in accordance with formulas (2), the Gram matrix of the basis $g_1, \dots, g_m$ has the shape $G = T \Gamma T^t$, where t stands for transposition. Since the Gram matrix of an orthogonal basis is diagonal, we arrive at the following assertion.

Lemma 1. *Let Γ be the Gram matrix of some basis for an integer m -dimensional lattice L . The lattice L possesses an orthogonal basis if and only if there exists a matrix $T \in GL_m(\mathbf{Z})$ such that $T \Gamma T^t$ is a diagonal matrix.*

We now demonstrate that construction of an orthogonal basis for a rational lattice reduces to construction of an orthogonal basis for some integer lattice. More precisely, the following assertion holds.

Lemma 2. *For every rational lattice L , there exists an isomorphic integer lattice M such that L possesses an orthogonal basis if and only if M possesses an orthogonal basis.*

PROOF. Observe that every nonzero rational vector a in V can be uniquely written down as

$$a = \frac{p}{q} (a^1, \dots, a^n), \quad (3)$$

where p and q are coprime natural numbers and all components $a^1, \dots, a^n$ are integers and their greatest common divisor is 1. Let a rational lattice $L = \mathcal{L}(a_1, \dots, a_m)$ be generated by rational vectors $a_1, \dots, a_m$. Write down each generating vector in the shape (3):

$$a_i = \frac{p_i}{q_i} (a_i^1, a_i^2, \dots, a_i^n), \quad p_i, q_i \in \mathbf{N}; \quad a_i^j \in \mathbf{Z}, \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

Denote the least common multiple of the denominators $q_1, \dots, q_m$ by d . Put $\bar{a}_i = da_i$, $i = 1, \dots, m$. Obviously, all vectors $\bar{a}_i$ are integer. As a sought lattice M we take the lattice $\mathcal{L}(\bar{a}_1, \dots, \bar{a}_m)$ which is isomorphic to L . As was established above, the lattice M possesses a ladder basis that consists of integer vectors $\bar{c}_1, \dots, \bar{c}_r$, $1 \leq r \leq m$. Then the vectors $c_i = \frac{1}{d}\bar{c}_i$, $i = 1, \dots, r$, constitute a basis for the lattice L . Basing on the properties of inner product, note that $c_i \cdot c_j = \frac{1}{d^2}(\bar{c}_i \cdot \bar{c}_j)$. Therefore, if Γ is the Gram matrix of the basis $\bar{c}_1, \dots, \bar{c}_k$ then the matrix $\frac{1}{d^2}\Gamma$ is the Gram matrix of the basis $c_1, \dots, c_k$. By virtue of Lemma 1, the lattices L and M do possess an orthogonal basis simultaneously. The proof of the lemma is over.

Henceforth, unless otherwise stated, we consider integer lattices with a fixed ladder basis.

The *support* $\text{supp}(x)$ of a vector $x = x^1e_1 + \dots + x^ne_n$ is the subset of those vectors $e_1, \dots, e_n$ for which the corresponding coordinates of x are nonzero. If $a_1, \dots, a_m$ is some set of vectors then the *support* $\text{supp}(a_1, \dots, a_m)$ of this system is the union of supports $\bigcup_{i=1}^m \text{supp}(a_i)$.

We formulate a sufficient condition for existence of an orthogonal basis for an integer lattice.

Lemma 3. *If, in a ladder basis $c_1, \dots, c_r$, $r \geq 2$, for an integer lattice L , the orthogonal projection of the vector c_i , $i = 1, \dots, r-1$, onto the support $\text{supp}(c_{i+1}, \dots, c_r)$ lies in the sublattice $\mathcal{L}(c_{i+1}, \dots, c_r)$, then the lattice L possesses an orthogonal basis.*

PROOF is performed by inducting on the dimension of the lattice L . By assumption, the vector $d^1e_1 + \dots + d^ne_n$, where the coefficient d^i equals c_i^1 if e_i lies in $\text{supp}(c_2, \dots, c_r)$ and equals zero otherwise, is the projection of c_1 onto $\text{supp}(c_2, \dots, c_r)$. By the hypothesis of the lemma, this vector belongs to the sublattice $\mathcal{L}(c_2, \dots, c_r)$. Hence, the following equality holds for some integers $\alpha_2, \dots, \alpha_r$:

$$d^1e_1 + \dots + d^ne_n = \alpha_2c_2 + \dots + \alpha_rc_r.$$

It is obvious that the vectors $f_1 = c_1 - (\alpha_2c_2 + \dots + \alpha_rc_r)$, $c_2, \dots, c_r$ constitute a basis for the lattice L and that the vector f_1 is orthogonal to the sublattice $\mathcal{L}(c_2, \dots, c_r)$. By the induction hypothesis, the lattice $\mathcal{L}(c_2, \dots, c_r)$ possesses an orthogonal basis $f_2, \dots, f_r$. Then $f_1, f_2, \dots, f_r$ constitute an orthogonal basis for L . The proof of the lemma is over.

We shall see below that the condition of the lemma is sufficient but not necessary for existence of an orthogonal basis.

§ 2. An Orthogonal Basis for a Two-Dimensional Lattice

In this section we consider two-dimensional integer lattices and establish a criterion for existence of an orthogonal basis for them.

Let L be a two-dimensional integer lattice generated by vectors u and v . By Lemma 1, L possesses an orthogonal basis if and only if there is a matrix

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL_2(\mathbb{Z})$$

such that the vector $\alpha u + \beta v$ is orthogonal to the vector $\gamma u + \delta v$. This condition amounts to the fact that the system of the equations

$$a\alpha\gamma + b(\alpha\delta + \beta\gamma) + c\beta\delta = 0, \quad \alpha\delta - \beta\gamma = 1 \quad (1)$$

in the unknowns α, β, γ , and δ is solvable over $\mathbb{Z}$, where a, b , and c are the entries of the Gram matrix

$$\Gamma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} u \cdot u & u \cdot v \\ u \cdot v & v \cdot v \end{pmatrix}$$

of the basis u, v .

We first suppose that system (1) possesses a solution $(\alpha, \beta, \gamma, \delta)$ which have zero components. It is clear that there may exist at most two zero components. If two elements among α, β, γ , and δ are

equal to zero then the vectors u and v are orthogonal and the entry b of Γ equals zero. If $\alpha = 0$ and the other components β , γ , and δ differ from zero then

$$T = \begin{pmatrix} 0 & -\varepsilon \\ \varepsilon & \lambda \end{pmatrix}, \quad \varepsilon = \pm 1, \quad \lambda \in \mathbf{Z} \setminus \{0\},$$

and the first equation in (1) looks like

$$b + \varepsilon\lambda c = 0.$$

If $\beta = 0$ and $\alpha\gamma\delta \neq 0$ then

$$T = \begin{pmatrix} \varepsilon & 0 \\ \lambda & \varepsilon \end{pmatrix}, \quad \varepsilon = \pm 1, \quad \lambda \in \mathbf{Z} \setminus \{0\},$$

and the first equation of the system has the shape

$$b + \varepsilon\lambda a = 0.$$

The cases in which $\gamma = 0$ and $\alpha\beta\delta \neq 0$ as well as $\delta = 0$ and $\alpha\beta\gamma \neq 0$ are settled similarly. This easily implies the following

Lemma 4. *If the entry b of the Gram matrix $\Gamma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ of a two-dimensional integer lattice L divides by a or c then L possesses an orthogonal basis.*

Henceforth we assume that all entries of Γ differ from zero and system (1) have no solutions $(\alpha, \beta, \gamma, \delta)$ such that $\alpha\beta\gamma\delta = 0$.

We proceed with solving system (1). Without loss of generality, we assume that the greatest common divisor of a , b , and c is 1. Inserting the expression for $\alpha\delta$ found from the second equation of the system in the first equation, we obtain

$$a\alpha\gamma + b(1 + 2\beta\gamma) + c\beta\delta = 0.$$

Put $d = \text{GCD}(b, c)$. Then $b = db_1$ and $c = dc_1$ for some coprime integers b_1 and c_1 . With these notations, the last equation takes the shape

$$a\alpha\gamma + d[b_1(1 + 2\beta\gamma) + c_1\beta\delta] = 0. \quad (2)$$

Since a and d are coprime, equation (2) is equivalent to the system

$$\alpha\gamma = -dt, \quad b_1(1 + 2\beta\gamma) + c_1\beta\delta = at, \quad (3)$$

where t is a nonzero integer. Rewrite the second equation of this system as

$$\beta(2b_1\gamma + c_1\delta) = at - b_1.$$

Suppose firstly that $2b_1\gamma + c_1\delta = 0$. Then $a|b_1$ and so $a|b$. In consequence, by Lemma 4 the lattice L possesses an orthogonal basis and, contrary to our assumption, system (1) has a solution $(\alpha, \beta, \gamma, \delta)$ for which $\alpha\beta\gamma\delta = 0$.

From now on, we assume that $2b_1\gamma + c_1\delta \neq 0$. Solve system (3) for α and β :

$$\alpha = -\frac{dt}{\gamma}, \quad \beta = \frac{at - b_1}{2b_1\gamma + c_1\delta}. \quad (4)$$

Inserting these expressions in the second equation of (1), after simple transformations we come to the equation

$$ct\delta^2 + (2bt + c_1)\gamma\delta + (at + b_1)\gamma^2 = 0.$$

The left-hand side of this equation can be regarded as a quadratic form in the variables δ and γ . Distinguishing a full square and eliminating denominators, we obtain

$$[2ct\delta + (2bt + c_1)\gamma]^2 + [4d(ac_1 - bb_1)t^2 - c_1^2]\gamma^2 = 0. \quad (5)$$

If the expression in the second brackets is nonnegative then (5) is possible if and only if both expressions in brackets vanish, i.e., when the system of the Diophantine equations

$$4d(ac_1 - bb_1)t^2 = c_1^2, \quad 2ct\delta + (2bt + c_1)\gamma = 0 \quad (6)$$

is solvable. For solvability of the first equation, it is necessary that the number $\det \Gamma = d(ac_1 - db_1^2) = ac - b_1^2$ be a square of some integer. If this condition is satisfied then

$$t = \pm \frac{c_1}{2\sqrt{ac - b^2}}$$

is an integer whenever $2\sqrt{ac - b^2} | c_1$. Assuming these conditions satisfied and considering the second equation of (6), observe that the fact that δ and γ are coprime implies that the equalities

$$2bt + c_1 = \delta t_1, \quad 2ct = -t_1\gamma$$

hold for some nonzero integer t_1 . From these equalities we infer that

$$\delta = \frac{2bt + c_1}{t_1}, \quad \gamma = -\frac{2ct}{t_1}. \quad (7)$$

Since δ and γ are coprime integers, $t_1 = \text{GCD}(2bt + c_1, 2ct)$. Inserting the expressions of (7) for δ and γ in (4), we obtain

$$\alpha = \frac{t_1}{2c_1}, \quad \beta = \frac{(at - b_1)t}{c_1^2 - 2bc_1t}.$$

Basing on the fact that α and β are nonzero integers and recalling the preceding relations, we arrive at the following assertion.

Lemma 5. *Suppose that the determinant of the Gram matrix*

$$\Gamma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} a & db_1 \\ db_1 & dc_1 \end{pmatrix}, \quad \text{GCD}(a, b, c) = 1, \quad d = \text{GCD}(b, c),$$

of a two-dimensional integer lattice L is a full square, i.e., $\det \Gamma = g^2$ for some positive integer g , and that the numbers $\frac{c_1}{2g}$, $\frac{t_1}{2c_1}$, and $\frac{(at - b_1)t}{c_1^2 - 2bc_1t}$, where $t_1 = \text{GCD}(2bt + c_1, 2ct)$, are nonzero integers for some $t = \pm \frac{c_1}{2g}$. Then the lattice L possesses an orthogonal basis.

We return to (5) and assume that the expression in the second brackets is negative. Then we can rewrite this equation as

$$[2ct\delta + (2bt + c_1)\gamma]^2 - [c_1^2 - 4d(ac_1 - bb_1)t^2]\gamma^2 = 0, \quad (8)$$

where the expression in the second brackets is now positive and, obviously, it must be a full square; i.e., the equality

$$c_1^2 - 4(ac - b^2)t^2 = \mu^2$$

holds for some integer μ . Denote the determinant of the Gram matrix by $g = \det \Gamma = ac - b^2$ and rewrite the last equation as

$$\mu^2 + g(2t)^2 = c_1^2. \quad (9)$$

All solutions of such equations are described in [9, p. 90]. Suppose that equation (9) is solvable and let (μ, t) be a pair of nonzero integers representing a solution. Then (8) takes the shape

$$[2ct\delta + (2bt + c_1 - \mu)\gamma][2ct\delta + (2bt + c_1 + \mu)\gamma] = 0.$$

Suppose that the first factor on the left-hand side vanishes, i.e.,

$$2ct\delta + (2bt + c_1 - \mu)\gamma = 0. \quad (10)$$

Put $d^- = \text{GCD}(2ct, 2bt + c_1 - \mu)$. Then the equalities $2ct = d^-r^-$ and $2bt + c_1 - \mu = d^-s^-$ hold for some coprime integers r^- and s^- . Since δ and γ are coprime, equation (10) has a solution $(\delta, \gamma) = \varepsilon(-s^-, r^-)$, $\varepsilon = \pm 1$. In view of (4), from here we obtain

$$\alpha = -\varepsilon \frac{dt}{r^-}, \quad \beta = \varepsilon \frac{(at - b_1)}{2b_1r^- - c_1s^-}.$$

Similarly, if the second factor vanishes, i.e.,

$$2ct\delta + (2bt + c_1 + \mu)\gamma = 0, \quad (11)$$

then we put $d^+ = \text{GCD}(2ct, 2bt + c_1 + \mu)$ and find coprime integers r^+ and s^+ for which $2ct = d^+r^+$ and $2bt + c_1 + \mu = d^+s^+$. Then equation (11) has a solution $(\delta, \gamma) = \varepsilon(-s^+, r^+)$, $\varepsilon = \pm 1$. Inserting these expressions in (4), we obtain

$$\alpha = -\varepsilon \frac{dt}{r^+}, \quad \beta = \varepsilon \frac{(at - b_1)}{2b_1r^+ - c_1s^+}.$$

Hence, the following assertion is valid.

Lemma 6. Suppose that, for the entries of the Gram matrix

$$\Gamma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} a & db_1 \\ db_1 & dc_1 \end{pmatrix}, \quad \text{GCD}(a, b, c) = 1, \quad d = \text{GCD}(b, c),$$

of a two-dimensional integer lattice L , the equation

$$\mu^2 + (ac - b^2)(2t)^2 = c_1^2$$

has a pair of integer solutions (μ, t) , $\mu t \neq 0$, such that the numbers

$$\frac{dt}{r}, \quad \frac{at - b_1}{2b_1r - c_1s},$$

where $2ct = d^0r$, $2bt + c_1 + \varepsilon\mu = d^0s$, $d^0 = \text{GCD}(2ct, 2bt + c_1 + \varepsilon\mu)$, $r, s \in \mathbf{Z}$, $\text{GCD}(r, s) = 1$, are nonzero integers for some $\varepsilon = \pm 1$. Then the lattice L possesses an orthogonal basis.

Thus, we have examined all possible cases in which system (1) is solvable. Since solvability of this system is equivalent to existence of an orthogonal basis for L , we obtain the following criterion.

Theorem 1. Let $L = \mathcal{L}(u, v)$ be a two-dimensional integer lattice and let

$$\Gamma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} a & db_1 \\ db_1 & dc_1 \end{pmatrix}, \quad \text{GCD}(a, b, c) = 1, \quad d = \text{GCD}(b, c),$$

be the corresponding Gram matrix. The lattice L possesses an orthogonal basis if and only if the entries of the matrix Γ satisfy at least one of the following conditions:

- (1) b divides by a or c ;
- (2) $\det \Gamma = g^2$ for some positive integer g , and the numbers $\frac{c_1}{2g}$, $\frac{t_1}{2c_1}$, and $\frac{(at-b_1)t_1}{c_1^2-2bc_1t}$, where $t_1 = \text{GCD}(2bt + c_1, 2ct)$, are nonzero integers for some $t = \pm \frac{c_1}{2g}$;
- (3) the equation $\mu^2 + (ac - b^2)(2\nu)^2 = c_1^2$ has a pair of integer solutions (μ, ν) , $\mu\nu \neq 0$, such that the numbers $\frac{d\nu}{r}$ and $\frac{a\nu-b_1}{2b_1r-c_1s}$, where $2c\nu = d^0r$, $2b\nu + c_1 + \varepsilon\mu = d^0s$, $d^0 = \text{GCD}(2c\nu, 2b\nu + c_1 + \varepsilon\mu)$, $r, s \in \mathbf{Z}$, $\text{GCD}(r, s) = 1$, are nonzero integers for some $\varepsilon = \pm 1$.

§ 3. An Orthogonal Basis for a Complete Two-Dimensional Lattice

We recall that a lattice L in a vector space V is said to be *complete* if its dimension equals the dimension of V . In this section we give a simpler criterion of existence of an orthogonal basis for a complete two-dimensional lattice L which leans upon the comparison between the bases for L and the integer matrices in $M_2(\mathbb{Z})$.

Suppose that integer vectors $a_1 = (a_1^1, a_1^2)$ and $a_2 = (a_2^1, a_2^2)$ form a basis for a complete lattice L . Then we can consider the matrix

$$A = \begin{pmatrix} a_1^1 & a_2^1 \\ a_1^2 & a_2^2 \end{pmatrix} \in M_2(\mathbb{Z})$$

as the matrix of a linear transformation taking the lattice $\mathbb{Z}e_1 \oplus \mathbb{Z}e_2$ onto the lattice L . To another basis for L , there corresponds the matrix TA , where T is a matrix in $GL_2(\mathbb{Z})$. If the matrix TA corresponds to an orthogonal basis for L then the matrix $TA(TA)^t$ is diagonal. The matrices satisfying this condition are described in the following obvious lemma.

Lemma 7. *Let $B \in M_2(\mathbb{Z})$ be a matrix such that BB^t is diagonal. Then B has the shape*

$$B = \begin{pmatrix} \varepsilon gx & -\varepsilon gy \\ fy & fx \end{pmatrix}, \quad \varepsilon = \pm 1, \quad g > 0, \quad f > 0, \quad x, y \in \mathbb{Z},$$

and $\text{GCD}(x, y) = 1$ whenever $xy \neq 0$.

Observe that if the matrix TA corresponds to an orthogonal basis for L then every matrix QTA , where Q is a matrix in the orthogonal group $O_2(\mathbb{Z})$, corresponds to an orthogonal basis for L as well. In particular, if there exists a matrix $T \in GL_2(\mathbb{Z})$ such that TA corresponds to an orthogonal basis then, without loss of generality, we may assume that $\det T = 1$.

Suppose that integer vectors u and v form a ladder basis for a complete two-dimensional lattice L . Then to this basis corresponds the triangular matrix

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in M_2(\mathbb{Z}). \quad (1)$$

Without loss of generality, we henceforth assume that the entries of the matrix A are coprime; i.e.,

$$\text{GCD}(a, b, c) = 1. \quad (2)$$

Moreover, by applying integer elementary transformations to the rows of A , we can achieve that the entries of A satisfy the inequalities

$$0 \leq b < c, \quad a > 0. \quad (3)$$

Clearly, if $b = 0$ then L possesses an orthogonal basis. A triangular integer matrix A satisfying conditions (1)–(3) is referred to as a *matrix of a standard basis*, and the basis for L corresponding to A is said to be a *standard basis*. It is obvious that every complete two-dimensional lattice L possesses a standard basis. The following assertion holds.

Lemma 8. *A standard basis for a complete two-dimensional integer lattice L is determined uniquely.*

PROOF. Let

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in M_2(\mathbb{Z})$$

be a matrix of a standard basis for a complete lattice L . Then to any other basis for L corresponds the matrix TA , where $T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in GL_2(\mathbb{Z})$. We put

$$B = TA = \begin{pmatrix} \alpha a & \alpha b + \beta c \\ \gamma a & \gamma b + \delta c \end{pmatrix}.$$

If we demonstrate that the matrix of a standard basis, constructed from B , coincides with A then everything will be proven. Since α and γ are coprime, by elementary transformations of the rows of B we reduce it to the shape

$$B_1 = \begin{pmatrix} a & b_1 \\ 0 & c_1 \end{pmatrix}, \quad b_1, c_1 \in \mathbf{Z}, \quad 0 \leq b_1 < c_1.$$

Since the determinant of A coincides with the determinant of B_1 to within sign, we have $c_1 = c$. Obviously, the matrices B and B_1 correspond to different bases for L . Therefore, we have $B_1 = T_1 B = T_1 T A$ for some matrix T_1 in $\text{GL}_2(\mathbf{Z})$. Hence,

$$T_1 T = B_1 A^{-1} = \begin{pmatrix} 1 & \frac{b_1 - b}{c} \\ 0 & 1 \end{pmatrix}.$$

However, the matrix $T_1 T$ has integer entries. Therefore, $b_1 - b = \lambda c$ for some integer λ ; i.e., $b_1 = \lambda c + b$. Since b_1 and b satisfy the inequalities $0 \leq b < c$ and $0 \leq b_1 < c$, this fact implies that $\lambda = 0$, and so $b_1 = b$, which finishes the proof of the lemma.

We now assume that some matrix of a standard basis for L is not diagonal. By Lemma 7, in this case the lattice L possesses an orthogonal basis if and only if L possesses a basis such that the corresponding matrix has the shape

$$B = \begin{pmatrix} gx & -gy \\ fy & fx \end{pmatrix} \in M_2(\mathbf{Z}), \quad g > 0, \quad f > 0, \quad x \neq 0, \quad y \neq 0 \in \mathbf{Z},$$

$$\text{GCD}(f, g) = \text{GCD}(x, y) = 1.$$

We construct a matrix of a standard basis from this matrix. We put $g_0 = \text{GCD}(g, y)$ and $f_0 = \text{GCD}(f, x)$. Then $g = g_0 g_1$ and $y = g_0 y_1$ for some coprime integers g_1 and y_1 , and $f = f_0 f_1$ and $x = f_0 x_1$ for some coprime integers x_1 and f_1 . Since g and f are coprime, so are g_0 and f_0 . Observe that $g_0 f_0 = \text{GCD}(gx, fy)$. To construct a matrix of a standard basis from B , we consider the matrix

$$B_0 = \begin{pmatrix} 0 & 0 \\ g_0 g_1 f_0 x_1 & -g_0^2 g_1 y_1 \\ f_0 f_1 g_0 y_1 & f_0^2 f_1 x_1 \end{pmatrix}$$

that is obtained from B by adding the zero row. Since $g_1 x_1$ and $f_1 y_1$ are coprime numbers, we have $\alpha g_1 x_1 + \beta f_1 y_1 = 1$ for some integers α and β . Multiply the second row of the matrix B_0 by α and the third by β . Adding the so-obtained rows to the first row, we come to the matrix

$$B_1 = \begin{pmatrix} g_0 f_0 & -\alpha g_0^2 g_1 y_1 + \beta f_0^2 f_1 x_1 \\ g_0 f_0 g_1 x_1 & -g_0^2 g_1 y_1 \\ g_0 f_0 f_1 y_1 & f_0^2 f_1 x_1 \end{pmatrix}.$$

Multiply the first row of this matrix by $-g_1 x_1$ and add the result to the second row. Afterwards, multiply the first row by $-f_1 y_1$ and add the result to the third row. Using the equality $\alpha g_1 x_1 + \beta f_1 y_1 = 1$, represent the so-obtained matrix as follows:

$$B_2 = \begin{pmatrix} g_0 f_0 & \beta f_0^2 f_1 x_1 - \alpha g_0^2 g_1 y_1 \\ 0 & -\beta f_1 g_1 (f_0^2 x_1^2 + g_0^2 y_1^2) \\ 0 & \alpha f_1 g_1 (f_0^2 x_1^2 + g_0^2 y_1^2) \end{pmatrix}.$$

Since α and β are coprime; using the Euclid algorithm, applying appropriate elementary transformations to the last two rows, and deleting the zero row, we arrive at the matrix

$$B_3 = \begin{pmatrix} g_0 f_0 & \beta f x - \alpha g y \\ 0 & f_1 g_1 (x^2 + y^2) \end{pmatrix}.$$

Observe that the diagonal entries of this matrix are positive. Choose an integer λ so that the difference $r = \beta f x - \alpha g y - \lambda f_1 g_1 (x^2 + y^2)$ satisfy the inequalities $0 \leq r < f_1 g_1 (x^2 + y^2)$. Then the matrix

$$B_4 = \begin{pmatrix} g_0 f_0 & r \\ 0 & f_1 g_1 (x^2 + y^2) \end{pmatrix}$$

is a matrix of a standard basis for the lattice L . In view of Lemma 8, the equality $B_4 = A$ holds, where the matrix A is determined by the conditions (1)–(3). Hence, the following system of equalities hold:

$$a = f_0 g_0, \quad c = f_1 g_1 (x^2 + y^2), \quad b + \lambda c = \beta f x - \alpha g y, \quad \alpha g_1 x_1 + \beta f_1 y_1 = 1, \quad \lambda \in \mathbb{Z}. \quad (4)$$

Thus, if the lattice L possesses an orthogonal basis then the entries of the corresponding matrix B satisfy (4). On the other hand, if there are positive integers f_0, g_0, f_1 , and g_1 and integers $x_1 \neq 0, y_1 \neq 0, \alpha, \beta$, and λ such that

$$\text{GCD}(g_1, y_1) = \text{GCD}(f_1, x_1) = \text{GCD}(f_0 x_1, g_0 y_1) = \text{GCD}(f_0 f_1, g_0 g_1) = 1, \quad (5)$$

and (4) holds then the matrix

$$B = \begin{pmatrix} g_0 g_1 f_0 x_1 & -g_0^2 g_1 y_1 \\ f_0 f_1 g_0 y_1 & f_0^2 f_1 x_1 \end{pmatrix}$$

determines an orthogonal basis for L . Indeed, there are only finitely many collections $(f_0, g_0, f_1, g_1, x_1, y_1)$ satisfying (5) together with the first two equations in (4). Take one of these collections and find integers α and β satisfying the last equation in (4). Next verify whether the number

$$\lambda = \frac{\beta f x - \alpha g y - b}{f_1 g_1 (x^2 + y^2)}$$

is integer. If not then consider another collection. If λ turns out to be integer then our collection is a solution. Clearly, α and β are determined from the last equation of (4) in a nonunique fashion. Let us demonstrate that if the number $\lambda_0 = \frac{\beta_0 f x - \alpha_0 g y - b}{f_1 g_1 (x^2 + y^2)}$ is integer for some solution (α_0, β_0) of the equation $\alpha g_1 x_1 + \beta f_1 y_1 = 1$, then the number $\lambda_1 = \frac{\beta_1 f x - \alpha_1 g y - b}{f_1 g_1 (x^2 + y^2)}$ is integer for any other solution (α_1, β_1) . Indeed, there exists an integer t such that $(\alpha_1, \beta_1) = (\alpha_0 - f_1 y_1 t, \beta_0 + g_1 x_1 t)$. Then $\lambda_1 = \lambda_0 + t$. Hence, if $\lambda_0 \in \mathbb{Z}$ then $\lambda_1 \in \mathbb{Z}$ as well. We have thus established the following criterion.

Theorem 2. *Let L be a complete two-dimensional lattice with the following matrix of a standard basis:*

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in M_2(\mathbb{Z}), \quad \text{GCD}(a, b, c) = 1, \quad 0 \leq b < c, \quad a > 0.$$

There is an orthogonal basis for L if and only $b = 0$ or b differs from zero and the following two conditions are satisfied:

(1) *there are coprime integers $f_0 > 0$ and $g_0 > 0$ such that $a = f_0 g_0$ and there are integers $f_1 > 0, g_1 > 0, x_1 \neq 0$, and $y_1 \neq 0$ such that $c = f_1 g_1 (f_0^2 x_1^2 + g_0^2 y_1^2)$ and*

$$\text{GCD}(f_1, g_1) = \text{GCD}(x_1, y_1) = \text{GCD}(g_1, y_1) = \text{GCD}(f_1, x_1) = 1;$$

(2) *for integers α and β satisfying the equation $\alpha g_1 x_1 + \beta f_1 y_1 = 1$, the coefficient c divides $\beta f_0^2 f_1 x_1 - \alpha g_0^2 g_1 y_1 - b$.*

Under these conditions, the lattice L possesses the orthogonal basis that consists of the vectors

$$(g_0 g_1 f_0 x_1, -g_0^2 g_1 y_1), \quad (f_0 f_1 g_0 y_1, f_0^2 f_1 x_1).$$

From this theorem we can easily infer a condition whose fulfillment implies that the lattice does not possess an orthogonal basis. More precisely, the following assertion holds.

Corollary. Let L be a complete two-dimensional lattice with the following matrix of a standard basis:

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in M_2(\mathbf{Z}), \quad 0 < b < c, \quad a > 0, \quad \text{GCD}(a, b, c) = 1.$$

If a , b , and c satisfy at least one of the conditions

- (1) c is free of squares and has as prime factors only numbers of the shape $4k + 3$;
- (2) $a \geq c$

then the lattice L does not possess an orthogonal basis.

PROOF. It is well known (see, for instance, [10, p. 116]) that an integer N is representable as the sum of two squares if and only if every prime factor of N of the shape $4k + 3$ enters into N with an even power. Hence, if c is free of squares and only has prime factors of the shape $4k + 3$, then c cannot be represented as $c = f_1 g_1 (f_0^2 x_1^2 + g_0^2 y_1^2)$, where the integers f_1 , g_1 , f_0 , g_0 , x_1 , and y_1 satisfy condition (1) of Theorem 2.

Suppose that condition (2) of the corollary is satisfied, but the lattice L nevertheless possesses an orthogonal basis. By Theorem 2, then there are positive integers f_0 and g_0 such that $a = f_0 g_0$ and there are integers f_1 , g_1 , x_1 , and y_1 such that $c = f_1 g_1 (f_0^2 x_1^2 + g_0^2 y_1^2)$. The obvious inequality $f_0^2 + g_0^2 + (f_0 - g_0)^2 > 0$ readily implies that $f_0^2 + g_0^2 > f_0 g_0$. Since $c \geq f_0^2 + g_0^2$, it follows that $c > a$, contradicting the supposition. The proof of the corollary is over.

§ 4. On Problem 9.45 in the "Kourovka Notebook"

Observe that Problem 9.45 cited in the introduction is a particular case of the problem of existence of an orthogonal basis for a rational lattice. Indeed, it is easy to see that $S(a)$ coincides with the lattice $\mathcal{L}(a, e_1, e_2, \dots, e_n)$ that is generated by the vectors $a, e_1, e_2, \dots, e_n$, where $e_1, e_2, \dots, e_n$ is an orthonormal basis for V .

We formulate a sufficient condition for existence of an orthogonal basis for the lattice $S(a)$.

Proposition 1. Let $a = (\frac{p_1}{q_1}, \dots, \frac{p_n}{q_n})$ be a rational vector in V , where all fractions $\frac{p_i}{q_i}$ are uncancellable and $q_j = 1$ for $p_j = 0$, $j = 1, \dots, n$. If all denominators $q_1, \dots, q_n$ are coprime then the lattice $S(a)$ possesses an orthogonal basis.

PROOF. Let us demonstrate that the lattice

$$S(a) = \left\{ \left(\frac{kp_1}{q_1} + b_1, \frac{kp_2}{q_2} + b_2, \dots, \frac{kp_n}{q_n} + b_n \right) \mid k, b_1, \dots, b_n \in \mathbf{Z} \right\} \quad (1)$$

contains all vectors

$$a_i = \left(0, \dots, 0, \frac{1}{q_i}, 0, \dots, 0 \right), \quad i = 1, \dots, n,$$

where the nonzero component occupies the i th position. Indeed, if $p_i = 0$ then $q_i = 1$, and it suffices to put in (1) $k = b_1 = \dots = b_{i-1} = b_{i+1} = \dots = b_n = 0$ and $b_i = 1$. Now, suppose that $p_i \neq 0$. We have to show that there are integers $k, b_1, \dots, b_n$ satisfying the system of the equations

$$kp_i + q_i b_i = 1, \quad kp_j + q_j b_j = 0, \quad j = 1, \dots, i-1, i+1, \dots, n. \quad (2)$$

Considering the equations of the system with the zero right-hand side, we see that

$$k = -\frac{q_j b_j}{p_j}.$$

Since k must be an integer, we put $b_j = -p_j b'_j$, where $b'_j \in \mathbf{Z}$ for all $j = 1, \dots, i-1, i+1, \dots, n$. Hence, we have the system of the equalities

$$k = q_1 b'_1 = q_2 b'_2 = \dots = q_{i-1} b'_{i-1} = q_{i+1} b'_{i+1} = \dots = q_n b'_n.$$

This system will hold if we put

$$b'_j = \frac{t}{q_i q_j} \prod_{l=1}^n q_l, \quad j = 1, \dots, i-1, i+1, \dots, n,$$

where t is an arbitrary integer. Therefore, $k = \frac{t}{q_i} \prod_{l=1}^n q_l$. Inserting this expression in the first equation of (2) and using the fact that the numbers $q_1 \dots q_{i-1} q_{i+1} \dots q_n p_i$ and q_i are coprime, we find integers t and b_i satisfying the first equation. Thus, all vectors a_i , $i = 1, \dots, n$, lie in $S(a)$. Furthermore, it is obvious that they are pairwise orthogonal and generate the whole lattice $S(a)$. So these vectors are sought ones and we finish the proof of the proposition.

Nevertheless, the condition of the proposition is not necessary for existence of an orthogonal basis, which is seen from the following

EXAMPLE. Take as a the vector $(-\frac{1}{6 \cdot 13}, \frac{3}{13})$. For it, the conditions of the proposition fail. However, it is easy to verify that the vectors $\frac{1}{26}(-3, 2)$ and $\frac{1}{39}(2, 3)$ form an orthogonal basis for the lattice $S(a)$.

From now on, we assume that $n = 2$. Suppose that $a = (\frac{p_1}{q_1}, \frac{p_2}{q_2})$, $q_1, q_2 \in \mathbb{N}$, and $p_1, p_2 \in \mathbb{Z}$. If $p_1 = 0$ then the vector $a' = (1, \frac{p_2}{q_2})$ determines the lattice $S(a')$ which coincides with the lattice $S(a)$. Indeed, since $S(a) = \mathcal{L}(a, e_1, e_2) = \mathcal{L}(a + e_1, e_1, e_2)$ and $a' = a + e_1$, we have $S(a) = S(a')$. Analogously, if $p_2 = 0$ then $S(a) = S(a'')$ for $a'' = (\frac{p_1}{q_1}, 1)$. Therefore, we can assume that both components of a differ from zero and that the fractions $\frac{p_1}{q_1}$ and $\frac{p_2}{q_2}$ are uncancellable. We now indicate some basis for the lattice $S(a)$. To this end, we find the greatest common divisor $q_0 = \text{GCD}(q_1, q_2)$. Then $q_i = q_0 q'_i$, $i = 1, 2$, for some coprime integers q'_1 and q'_2 . Represent the vector a as

$$a = \frac{1}{q_0 q'_1 q'_2} (p_1 q'_2, p_2 q'_1)$$

and consider the matrix

$$A = \begin{pmatrix} p_1 q'_2 & p_2 q'_1 \\ q_0 q'_1 q'_2 & 0 \\ 0 & q_0 q'_1 q'_2 \end{pmatrix}$$

which generates the integer lattice corresponding to the lattice $S(a)$. Find a basis for this integer lattice. To this end, choose integers u and v such that

$$p_1 u + q_1 v = 1. \quad (3)$$

Multiply the first row of A by u and add the result to the third row. Afterwards, multiply the second row by v and add the result to the third row. We obtain the matrix

$$\begin{pmatrix} p_1 q'_2 & p_2 q'_1 \\ q_1 q'_2 & 0 \\ q'_2 & q'_1(q_2 + u p_2) \end{pmatrix}.$$

Multiply the last row of this matrix by $-q_1$ and add the result to the second row, next multiply the last row by $-p_1$ and add the result to the first row. Interchanging the first and third rows in the resultant matrix, we obtain

$$\begin{pmatrix} q'_2 & (q_2 + u p_2) q'_1 \\ 0 & -q_1(q_2 + u p_2) q'_1 \\ 0 & (v q_1 p_2 - p_1 q_2) q'_1 \end{pmatrix}. \quad (4)$$

Consider the last two entries of the second column of this matrix and demonstrate that their greatest common divisor equals q_1 . To this end, it suffices to show that $\text{GCD}(q'_1(q_2 + u p_2), v q'_1 p_2 - p_1 q'_2) = 1$. It is obvious that $\text{GCD}(q'_1, v q'_1 p_2 - p_1 q'_2) = 1$. Put $d = \text{GCD}(q_2 + u p_2, v q'_1 p_2 - p_1 q'_2)$. Then each of

the elements $q_2 + up_2$ and $q_0(vq'_1p_2 - p_1q'_2) = vq_1p_2 - p_1q_2$ is divisible by d . Put $l = q_2 + up_2$. Then $q_2 = l - up_2$ and $vq_1p_2 - p_1q_2 = vq_1p_2 - lp_1 + up_1p_2$. Hence, $d|p_2(vq_1 + up_1)$. In view of (3), this means however that $d|p_2$. On the other hand, $d|(q_2 + up_2)$. Thereby $d|q_2$. Since p_2 and q_2 are coprime, this means that $d = 1$. Using the Euclid algorithm and applying appropriate elementary transformations to the last two rows of the matrix (4), we obtain the matrix

$$B = \begin{pmatrix} q'_2 & (q_2 + up_2)q'_1 \\ 0 & q_1 \end{pmatrix}. \quad (5)$$

Denote by $R(a)$ the lattice that is generated by the rows of this matrix. It is clear that $R(a)$ coincides with the lattice generated by the rows of the matrix A . Since the vectors

$$b_1 = \left(\frac{1}{q_1}, \frac{q_2 + up_2}{q_2} \right), \quad b_2 = \left(0, \frac{1}{q_2} \right)$$

form a basis for the lattice $S(a)$, it is obvious that the lattices $R(a)$ and $S(a)$ possess an orthogonal basis simultaneously. Thereby we have established the following

Lemma 9. Suppose that $a = \left(\frac{p_1}{q_1}, \frac{p_2}{q_2} \right)$, $q_1, q_2 \in \mathbb{N}$, $p_1, p_2 \in \mathbb{Z}$, and the fraction $(p_i, q_i) = 1$, $i = 1, 2$, is uncancellable. Then the lattice $S(a)$ possesses the basis

$$b_1 = \left(\frac{1}{q_1}, \frac{q_2 + up_2}{q_2} \right), \quad b_2 = \left(0, \frac{1}{q'_2} \right), \quad q'_2 = \frac{q_2}{\text{GCD}(q_1, q_2)},$$

where u satisfies the equation $p_1u + q_1v = 1$.

Lemma 10. For every vector $a = \left(\frac{p_1}{q_1}, \frac{p_2}{q_2} \right) \in \mathbb{Q}^2$, there is an integer p such that the lattice $S(a_0)$ for the vector $a_0 = \left(\frac{1}{q_1}, \frac{p}{q_2} \right)$ coincides with the lattice $S(a)$.

PROOF. Since $S(a) = \mathcal{L}(a, e_1, e_2)$, it is obvious that $S(a) = S(a + e_i)$, where $i \in \{1, 2\}$. Hence, if $p_1 = 0$ then, putting $p = p_2$, we obtain $S(a) = S(a + e_1) = S(a_0)$. Therefore, from now on we assume that both components of a differ from zero. By the proof of Lemma 9, to the lattice $S(a)$ corresponds the integer lattice $R(a)$ with the matrix B determined by (5). It is easy to see from here that the lattice $R(a)$ is generated by any pair of vectors (c, c_k) , $k \in \mathbb{Z}$, where

$$c = (0, q_1), \quad c_k = (q'_2, (q_2 + up_2 + q_0k)q'_1), \quad k \in \mathbb{Z}.$$

Consider the arithmetic progression

$$q_2 + up_2, \quad q_2 + up_2 + q_0, \quad q_2 + up_2 + 2q_0, \dots$$

with the first member $q_2 + up_2$ and difference q_0 . Since q_0 and $q_2 + up_2$ are coprime, by the Dirichlet theorem there are infinitely many prime numbers in this progression. Find the least nonnegative number $k_0 \in \mathbb{Z}$ such that $q_2 + up_2 + q_0k_0$ and q_2 are coprime numbers. Put $p = q_2 + up_2 + q_0k_0$. Demonstrate that the so-constructed number p possesses all needed properties. Indeed, by construction the fraction $\frac{p}{q_2}$ is uncancellable. We must next show that the lattices $S(a)$ and $S(a_0)$, where $a_0 = \left(\frac{1}{q_1}, \frac{p}{q_2} \right)$, coincide. To this end, it suffices to prove coincidence of the integer lattices $R(a)$ and $R(a_0)$ corresponding to $S(a)$ and $S(a_0)$. Arguing like in Lemma 9, we construct the matrix

$$\begin{pmatrix} q'_2 & (q_2 + u_1p)q'_1 \\ 0 & q_1 \end{pmatrix}, \quad u_1 + q_1v_1 = 1,$$

for the lattice $R(a_0)$. Inserting $u_1 = 1 - q_1v_1$, we obtain the matrix

$$\begin{pmatrix} q'_2 & pq'_1 + q_1(q'_2 - pv_1q'_1) \\ 0 & q_1 \end{pmatrix}.$$

Multiply the second row by $pv_1q'_1 - q'_2$ and add the result to the first row to obtain the matrix

$$B_0 = \begin{pmatrix} q'_2 & pq'_1 \\ 0 & q_1 \end{pmatrix}$$

which corresponds to a basis for the lattice $R(a_0)$. Hence, the pair of the vectors c, c_{k_0} forms a basis for the lattice $R(a_0)$. Since these vectors are basic vectors for the lattice $R(a)$ too, the required assertion follows. The proof of the lemma is over.

In view of this lemma, we can assume that the vector a is as follows: $a = (\frac{1}{q_1}, \frac{p_2}{q_2})$, $p_2 \neq 0$. Then to the lattice $S(a)$ corresponds the integer lattice $R(a)$ with the matrix

$$B_0 = \begin{pmatrix} q'_2 & p_2q'_1 \\ 0 & q_1 \end{pmatrix}.$$

Choose an integer λ so that the sum $p_2q'_1 + \lambda q_1$ satisfy the inequalities $0 \leq p_2q'_1 + \lambda q_1 < q_1$. Then the matrix

$$\begin{pmatrix} q'_2 & p_2q'_1 + \lambda q_1 \\ 0 & q_1 \end{pmatrix}$$

is a matrix of a standard basis for the lattice $R(a)$. This matrix is diagonal if and only if $q_0 = \text{GCD}(q_1, q_2) = 1$. By Lemma 2, the lattice $S(a)$ possesses an orthogonal basis if and only if the lattice $R(a)$ possesses an orthogonal basis. Therefore, from Theorem 2 we immediately obtain the following

Proposition 2. *Let $S(a)$ be the lattice determined by the vector $a = (\frac{1}{q_1}, \frac{p_2}{q_2})$, $q_1, q_2 \in \mathbb{N}$, $p_2 \in \mathbb{Z} \setminus \{0\}$, $\text{GCD}(p_2, q_2) = 1$. The lattice $S(a)$ possesses an orthogonal basis if and only if q_1 and q_2 are coprime or they are not coprime but the following two conditions are satisfied:*

(1) *if $q_0 = \text{GCD}(q_1, q_2)$, $q'_i = \frac{q_i}{q_0}$, $i = 1, 2$, then there are coprime integers f_0 and g_0 and nonzero integers $f_1 > 0$, $g_1 > 0$, x_1 , and y_1 such that $q'_2 = f_0g_0$, $q_1 = f_1g_1(f_0^2x_1^2 + g_0^2y_1^2)$, and $\text{GCD}(g_1, f_1) = \text{GCD}(f_1, x_1) = \text{GCD}(g_1, y_1) = \text{GCD}(x_1, y_1) = 1$;*

(2) *for integers α and β satisfying the equation $\alpha g_1x_1 + \beta f_1y_1 = 1$, the coefficient q_1 divides $\beta f_0^2f_1x_1 - \alpha g_0^2g_1y_1 - p_2q'_1 - \lambda q_1$, where λ is an integer such that $0 \leq p_2q'_1 + \lambda q_1 < q_1$.*

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